



MCAST

Hybrid Vision-Based Traffic Anomaly Detection Using Vehicle Tracking and Speed Estimation

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A dissertation submitted to the Institute of Information and Communication Technology in partial fulfilment of the requirements for the degree of BSc (Hons) Multimedia in Software Development

Authorship Statement

This dissertation is based on the results of research carried out by myself, is my own composition, and has not been previously presented for any other certified or uncertified qualification.

The research was carried out under the supervision of Marco Farrugia

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Date

.....

Signature

Copyright Statement

In submitting this dissertation to the MCAST Institute of Information and Communication Technology, I understand that I am giving permission for it to be made available for use in accordance with the regulations of MCAST and the Library and Learning Resource Centre. I accept that my dissertation may be made publicly available at MCAST's discretion.

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Date

.....

Signature

Acknowledgements

The list of people that the Student would like to thank on the completion of the dissertation. Mr Marco Farrugia, who supported me during my dissertation work as my tutor.

Abstract

This section should clearly state what the study is about, summarizing how it was carried out and what the results were. References are not to be included in the abstract. It should present only the essentials of the work in general.

Keywords: Dissertation, keywords.

Table of Contents

Authorship Statement	i
Copyright Statement	ii
Acknowledgements	iii
Abstract	iv
List of Figures	vii
List of Tables	viii
List of Abbreviations	ix
1 Introduction	1
1.1 Sub-chapter One	1
1.2 Sub-chapter Two	3
1.3 Sub-chapter Three	4
1.4 Sub-chapter Four	5
1.5 Sub-chapter Five	6
2 Literature Review	8
2.1 Overview	8
2.2 One	8
2.3 Two	9
2.4 Three	10
2.4.1 Four	11
2.4.2 Five	11
3 Research Methodology	13
3.1 One	13
3.2 Two	13
3.3 Three	13
4 Analysis of Results and Discussion	14
4.1 One	14
4.2 Two	14
4.3 Three	14
5 Conclusions and Recommendations	15
5.1 One	15
5.2 Two	15

5.3 Three	15
List of References	16
Appendix A Introduction of Appendix	17
Appendix B Sample Code	18

List of Figures

1.1	Create a Booktabs style table	2
1.2	Bounding-box example of cars.	3
2.1	Machine Learning.	9

List of Tables

1.1	Age of Participants	1
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List of Abbreviations

NN	Neural Network
ML	Machine Learning
DL	Deep Learning
FCN	Fully Convolutional Network
CNN	Convolutional Neural Network

Chapter 1: Introduction

In this section, **you**, the Student, are expected to state clearly:

- (a) the ‘problem’ or ‘question’ being researched;
- (b) why this topic was chosen;
- (c) what motivated the you to choose this topic;
- (d) why did you investigate the topic the way you did;
- (e) what problem did the you wish to explore;
- (f) what is the context for the research?

Percentage amount of words in section: 10 % of Dissertation*

1.1 Sub-chapter One

Background goes here. Also you can put in some references [1].

Another example of citations [1–3].

Here is a sample of table in Table 1.1

Age Groups	Frequency	Percent	Valid Percent	Cumulative Percent
16-20 years	100	98.2	98.2	95.2
21-25 years	5	4.8	4.8	100.0
Total	105	100.0	100.0	

Table 1.1: Age of Participants

Use \newpage to force start a new page.

A very quick way to create tables in a point & click environment is to use an online table generator¹

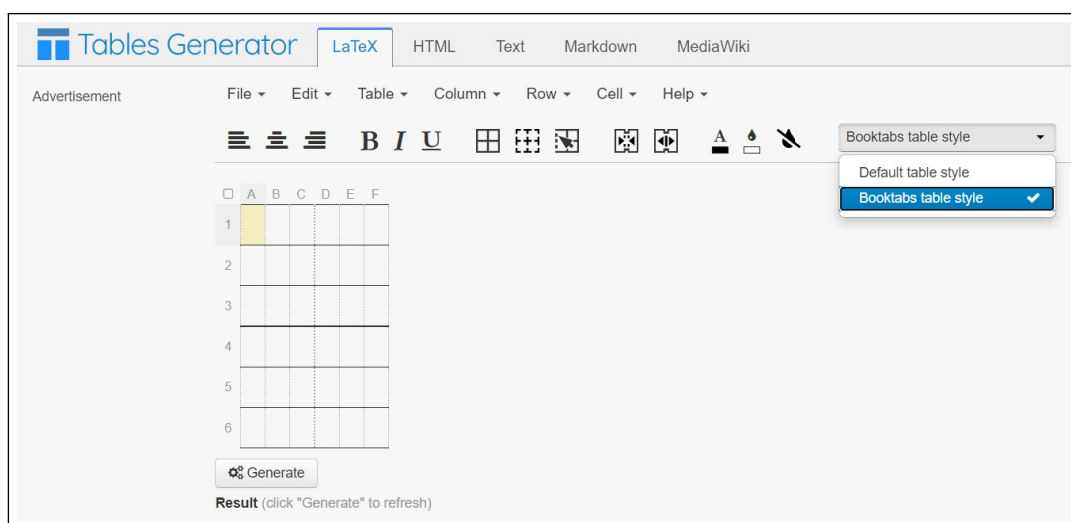


Figure 1.1: Create a Booktabs style table

Use `\enquote` for double-quotes. “This is a sample quote.”

Also can try to refer to this image in Figure 1.2. Notice that the `.eps` and `.pdf` format vector graphs are favoured, because:

1. they can be zoomed-in to check the detail.
2. text in such formats are search-able.

Try to insert a math equation as in Equation 1.1. If you wanna try the in-line mathematical, here is a sample $\alpha = \pi \cdot \frac{1}{\Theta}$.

$$e^{ix} = \cos x + i \sin x \quad (1.1)$$

When mention some file formats can use `music.mp3`, `latex.pdf`, etc.

¹<https://www.tablesgenerator.com>

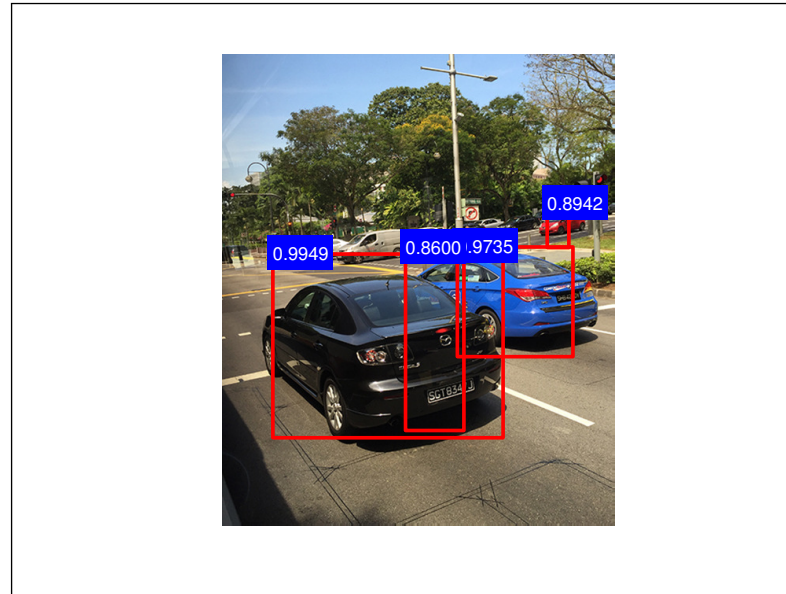


Figure 1.2: Bounding-box example of cars.

1.2 Sub-chapter Two

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of de Finibus Bonorum et Malorum (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, “Lorem ipsum dolor sit amet”..., comes from a line in section 1.10.32.

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum

This is a todo
note which
appears in the
margin

is that it has a more-or-less normal distribution of letters, as opposed to using 'Content here, content here', making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for 'lorem ipsum' will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

1.3 Sub-chapter Three

This todo note appears in line with the text. Todo notes are a great tool to leave comments or notes for self and can then be simply commented out.

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1.4 Sub-chapter Four

To create a bulleted list you need to make use of the "itemize" environment. A $\text{\LaTeX}$ environment is a special section within a page where items are placed. An environment always has a " $\backslash$ begin" and a " $\backslash$ end". Items you can place in such an environment include:

- enumerate – used to create numbered/letter lists
- itemize – creates a bullet list
- figures – to add images or figures (.eps is the recommended format)
- tables – you can create them in tablesgenerator.com then copy the $\text{\LaTeX}$ code and paste it.
- equations – for those really neat looking mathematical formulae
- this list is not exhaustive...

1.5 Sub-chapter Five

The following are some of the most common commands used during your writing. Some characters are reserved and cannot be directly written as in a normal word processor but need to be “escaped”. Then there are other useful commands such as adding a footnote, inserting a new line (to begin a new paragraph after a blank line, adding labels to items for cross-referencing, etc.

- The following are special characters which require a `\` before each character so the character is reproduced on the output. The tilde and exponent are even more special.

- `\#`
- `\$`
- `\%` – comment
- `\&`
- `\_`
- `~` – type `\textasciitilde`
- `^` – type `\textasciicircum`

- `\textbf` – bold font
- `\textit` – italics
- `\underline{words to underline}`

- `\emph` – emphasized text. If used within an italicized text, the output is normal font. If used by itself, text in its argument is italicized.

An example highlighting the use of `\emph`.

- An example of emphasized text with *these words emphasized* within normal font.
- *This is an example where these three words are emphasized within italicized font*

Chapter 2: Literature Review

2.1 Overview

The main purpose of a literature review is to show the reader that the Student studied and analyzed viewpoints of other researchers on the problem under consideration. A literature review is not just a summary of the books read but rather a thorough analysis of other viewpoints on the problem being analysed. Percentage amount of words in section: 25 % of Dissertation.

2.2 One

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Figure 2.1: Machine Learning.

2.3 Two

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2.4.2 Five

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Chapter 3: Research Methodology

3.1 One

It presents the chosen research methods and explains why these methods are effective. Percentage amount of words in section: 20 % of Dissertation.

3.2 Two

3.3 Three

Chapter 4: Analysis of Results and Discussion

4.1 One

This section includes critical discussion about the Student's findings and shows how these findings support the original objectives laid out for the dissertation, which may be partially or fully achieved, or even exceeded. The Student may also include new areas of an investigation prompted by developments in the research dissertation. Above all, it is required to present strong arguments which show how findings may offer a valid contribution to the development of the subject of the selected research area or issues related to it. Percentage amount of words in section: 25 % of Dissertation

4.2 Two

4.3 Three

Chapter 5: Conclusions and Recommendations

5.1 One

In this chapter, the Student has to evaluate the significance of the work done and give recommendations for any further investigations. Percentage amount of words in section: 20 % of Dissertation.

5.2 Two

5.3 Three

List of References

- [1] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” in *International Conference on Medical image computing and computer-assisted intervention*. Springer, 2015, pp. 234–241.
- [2] L. Lamport, *LaTeX: a Document Preparation System*, 2nd ed. Massachusetts: Addison Wesley, 1994.
- [3] J. Smith and J. Doe, “Deep learning for image classification,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 31, no. 5, pp. 1234–1245, 2020, doi: 10.1109/TNNLS.2020.2971234.

Chapter A: Introduction of Appendix

Interview summaries, sample questionnaires, and references should be placed in this section. For easier referencing, figures, tables, graphs, photos, diagrams, etc., should be inserted within the main text such as the literature review, the experimental process or procedure, the results and discussion chapters. Appendices are usually used to present further details about the results. Appendices may be a compulsory part of a dissertation, but they are not treated as part of the dissertation for purposes of assessing the dissertation. So any material which is significant to judging the quality of the dissertation or of the project as a whole should be in the main body of the dissertation (main text), and not in appendices.

Chapter B: Sample Code

You can share your GitHub link. Below shows how to insert highlighted source code from the source file.

```
# This function takes an integer n as input and returns the sum of numbers from 1 to n
def find_sum(n):
    sum = 0
    for i in range(1, n+1):
        sum += i
    return sum

# Ask the user for input and call the function
n = int(input("Enter an integer: "))
result = find_sum(n)

# Output the result
print("The sum of numbers from 1 to", n, "is", result)
```